

Adepu Nikhil

Hyderabad, Telangana, India

+91 7981340358

adepunikhil79@gmail.com

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To

Prof. Debabrata Maiti

DM Machine Learning Lab (CMTMC)

Department of Chemistry & CMTMC

Indian Institute of Technology Bombay

Mumbai, Maharashtra, India

Subject: Expression of Interest for Research Internship / Research Trainee Opportunity

Dear Prof. Maiti,

I hope you are doing well.

I am writing to express my sincere interest in the research activities of the DM Machine Learning Lab (CMTMC), IIT Bombay and to seek an opportunity to contribute as a Research Intern / Research Trainee under your guidance.

I have recently completed my **Master of Science in Artificial Intelligence and Data Science** from the **Central University of Andhra Pradesh**. My academic background and research interests are **centered on machine learning, deep learning, computer vision, generative AI, and intelligent real-time systems**. During my master's dissertation, I developed a **Distance-Aware Real-Time Assistive Navigation System using YOLOv8 and Monocular Depth Estimation**, where I worked on real-time object detection, spatial reasoning, FastAPI-based deployment, and multilingual feedback mechanisms.

I have also gained practical experience through internships in AI and data science, involving the development of machine learning models, computer vision pipelines, and AI-enabled applications using **Python, Scikit-learn, OpenCV, YOLOv8, FastAPI, and Generative AI tools**. In addition, I have contributed to a peer-reviewed publication related to computer vision and intelligent navigation systems, which strengthened my interest in research methodology, experimentation, and scientific problem solving.

I am particularly motivated by the research environment at IIT Bombay and would be honored to contribute to ongoing projects involving Machine Learning in Chemistry, Catalysis & Scientific Computing, AI for Science. My background in machine learning models, computer vision pipelines, and system integration aligns closely with your laboratory's focus, and I am eager to apply my technical skills to support your ongoing research initiatives.

I have attached my resume for your kind consideration. I would be grateful if my profile could be considered for any current or upcoming research opportunities in your laboratory.

Thank you very much for your time and consideration. I look forward to the possibility of working with your research group.

Sincerely,

Adepu Nikhil